

Dong-Kyum Kim

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SUMMARY

I am studying how AI systems store, retrieve, and update their knowledge. My research looks inside large language models to understand how they represent information and what can go wrong when we try to edit or remove knowledge from them. My background is in statistical physics and neuroscience, and I use ideas from learning and memory research in the brain to study how these systems work internally. I believe that understanding internal information processing in AI systems is essential for making them safer and more controllable.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) 2016–2022
Ph.D. in Physics Advisor: Prof. [Hawoong Jeong](#)
Dissertation: *Nonequilibrium Statistical Physics Study using Deep Learning*

Seoul National University (SNU) 2011–2015
B.S. in Physics, Minor in Computer Science & Engineering

RESEARCH EXPERIENCE

Postdoctoral Researcher, Max Planck Institute for Security and Privacy (MPI-SP) Oct. 2024–Present
• Hosted by Prof. [Meeyoung Cha](#) (Director)
• Data Science for Humanity Group

Postdoctoral Researcher, Institute for Basic Science (IBS) Mar. 2022–Sep. 2024
• Hosted by Prof. [Meeyoung Cha](#) (Chief Investigator)
• Data Science Group, Center for Mathematical and Computational Science

Data Science Intern, Samsung Electronics Sep.–Dec. 2017
• Collaborated with [Daniel Kim](#) (Senior Data Scientist)
• Improved anomaly image classification via distributed multi-GPU training (Keras & Spark)
• Implemented a distributed image searching framework via Elasticsearch

SELECTED PUBLICATIONS (INTERPRETABILITY-RELEVANT)

- [1] Jea Kwon, **Dong-Kyum Kim**, Jiwon Kim, Yonghyun Kim, Woong Kook, and Meeyoung Cha. “AI Engram: In Search of Memory Traces in Artificial Intelligence.” *ICML 2026 (Spotlight)*.
Geometric framework formalizing neuroscientific engram criteria; closed-form estimator isolates individual memory traces, enabling surgical composition or erasure of learned knowledge from MLPs to LLMs.
- [2] **Dong-Kyum Kim**, Minsung Kim, Jea Kwon, Nakyeong Yang, and Meeyoung Cha. “[Bilinear representation mitigates reversal curse and enables consistent model editing](#).” *ICLR 2026*.
Probed LM internal representations to discover bilinear geometric structure; enables logically consistent model editing.
- [3] Nakyeong Yang, **Dong-Kyum Kim**, Jea Kwon, Minsung Kim, Kyomin Jung, and Meeyoung Cha. “[Erase or Hide? Suppressing Spurious Unlearning Neurons for Robust Unlearning](#).” *ICLR 2026*.
Neuron-level mechanistic analysis of machine unlearning; identifies spurious neurons that mask rather than erase knowledge.
- [4] **Dong-Kyum Kim**[†], Jea Kwon[†], Meeyoung Cha, and C. Justin Lee. “[Transformer as a hippocampal memory consolidation model based on NMDAR-inspired nonlinearity](#).” *NeurIPS 2023*.
Interdisciplinary collaboration with electrophysiologist; reverse-engineered transformer memory via hippocampal consolidation theory.

ADDITIONAL PUBLICATIONS

- [5] Minsung Kim, **Dong-Kyum Kim**, Jea Kwon, Nakyeong Yang, Kyomin Jung, and Meeyoung Cha. “How Training Data Shapes the Use of Parametric and In-Context Knowledge in Language Models.” *ACL* 2026.
- [6] Minsung Kim, Jea Kwon, **Dong-Kyum Kim**, and Meeyoung Cha. “In Search of the Engram in LLMs.” *ICLR 2025 Blogpost*.
- [7] Gwangsu Kim, **Dong-Kyum Kim**, and Hawoong Jeong. “Spontaneous emergence of rudimentary music detectors in deep neural networks.” *Nature Communications* 2024.
- [8] Jea Kwon, Sunpil Kim, **Dong-Kyum Kim**, Jinhyeong Joo, SoHyung Kim, Meeyoung Cha, and C. Justin Lee. “SUBTLE: An unsupervised platform with temporal link embedding that maps animal behavior.” *IJCV* (2024).
- [9] Sangyun Lee, **Dong-Kyum Kim**, Jong-Min Park, Won Kyu Kim, Hyunggyu Park, and Jae Sung Lee. “Multidimensional entropic bound.” *Phys. Rev. Research* **5**, 013194 (2023).
- [10] **Dong-Kyum Kim**[†], Sangyun Lee[†], and Hawoong Jeong. “Estimating entropy production with odd-parity state variables via ML.” *Phys. Rev. Research* **4**, 023051 (2022).
- [11] Youngkyoung Bae, **Dong-Kyum Kim**, and Hawoong Jeong. “Inferring dissipation maps from videos using CNNs.” *Phys. Rev. Research* **4**, 033094 (2022).
- [12] **Dong-Kyum Kim** and Hawoong Jeong. “Deep reinforcement learning for feedback control in a collective flashing ratchet.” *Phys. Rev. Research* **3**, L022002 (2021).
- [13] **Dong-Kyum Kim**[†], Youngkyoung Bae[†], Sangyun Lee, and Hawoong Jeong. “Learning Entropy Production via Neural Networks.” *Phys. Rev. Lett.* **125**, 140604 (2020).
- [14] **Dong-Kyum Kim**[†], Byunghwee Lee[†], Daniel Kim, and Hawoong Jeong. “Multi-label classification of historical documents using hierarchical attention.” *J. Korean Phys. Soc.* **76**, 368 (2020).

TECHNICAL SKILLS

- **Daily driver:** Currently running multiple Claude Code sessions (Max plan, \$200/mo) via tmux as my primary research workflow.
- **ML/Interp:** PyTorch, HuggingFace Transformers, TransformerLens, nnsight, model probing & activation extraction, model editing (ROME/MEMIT)
- **Infrastructure:** Multi-GPU training & inference (Slurm clusters, hundreds of A100s), distributed training, CUDA, large-scale data pipelines
- **Languages:** Python (primary), C/C++, Bash
- **Other:** Git, Docker, Linux, Elasticsearch, Spark

ACADEMIC SERVICE

Reviewer: ICML 2026, ICLR 2026, NeurIPS 2025, ICML 2025 Workshop, ICLR 2025 Workshop, NeurIPS 2024 Workshop, ICML 2024 Workshop

AWARDS

Pre-doctoral Fellow of Physics, KAIST

2021

REFERENCES

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